

RECONNAISSANCE.md

EFT Codebase Reconnaissance — Honest Audit

The Mirror Platform LLC · April 2026

This report inventories the EFT codebase as it actually exists on disk, identifies divergences from the handoff documents, verifies (or falsifies) each architectural correction in handoff §17, evaluates D-1 Revision 2 testability, examines DEAP integration, and lists open implementation gaps in priority order.

The report distinguishes “implemented” (the function processes inputs to produce outputs through actual computation) from “scaffolded” (the function accepts inputs and returns outputs but the inside is a lookup table, a hand-typed value, or a passthrough). This distinction is non-negotiable.

When the documents and the code disagree, the code wins.

SECTION 1 — File-by-File Honest Inventory

`eft_verified_core.py` — 276 lines

Module-level constants (lines 13–31): `DIM=5`, `GHOST_FLOOR=0.05`, `DEPOSIT_THRESHOLD=0.12`, `LAMBDA_DECAY=15.0`, `KAPPA=0.3`, `BETA=1.0`, `SIGMA_MIN/MAX=0.3/1.5`, `S_THRESH=0.15`, `T_SIG=0.1`, `B_DIM=3`, `H_DIM=DIM`, `GAMMA_BODY=0.1`, `ALIGNMENT_REGEN=0.8`. Eigenmode indices `V_T=0`, `V_A=1`, `V_B=2`, `V_R=3`, `V_E=4`, `D4_INDICES=[0,1,2,3]`, `EXIT_INDEX=4`. **All implemented.**

BodySubstrate (lines 33–66): Maintains `b_star` (homeostatic baseline), current `b`, and `w_phi` projection matrix (3 rows × 5 columns, hand-typed coefficients on lines 39–43). Methods `phi`, `step` (homeostatic restoration $-\Gamma \cdot (b - b^*) + \xi$), `h`, `G_field` (`= expm(diag(h))`), `G_body`. **Implemented.** The `w_phi` matrix entries are hand-assigned literals — not learned, not derived. This is the body-state-to-affective-coordinate map and it is scaffolding from the corpus’s perspective.

sigma_from_salience (lines 68–70): Sigmoid mapping `s` → kernel width in `[SIGMA_MIN, SIGMA_MAX]`. **Implemented.**

Trace dataclass (lines 72–89): `position`, `significance`, `kernel_width`, `normal`, `deposit_time`,

`current_age`. Methods `persistence` ($= \exp(-\tau/\lambda)$), `effective_significance` ($= \max(s \cdot \text{persistence}, \text{GHOST_FLOOR})$) — **the ghost floor is enforced here**, line 84. Method `kernel_at` (Gaussian). **Implemented.**

`build_s_traces` (lines 91–98): Sums $-s_{\text{eff}} \cdot K \cdot (n \otimes n)$ across all traces.

Implemented. This is V2 sign convention — see Section 3.

`build_G_total` (lines 100–119): $\expm(w_f \cdot \log(G_{\text{field}}) + w_b \cdot \log(G_{\text{body}}) + w_t \cdot S_{\text{traces})$ with positive-definiteness guard at line 118 (clips minimum eigenvalue).

Implemented.

`psi` (lines 121–126): $-\log \det(G_{\text{total}})$ with sign-degenerate fallback to $1e10$.

Implemented.

`find_R_star` (lines 128–152): Gradient descent on Ψ with finite differences, weighted-trace-centroid initialization. **Implemented.** Returns single attractor — no secondary structure. See Section 3.6.

`compute_Hessian_at_R_star` (lines 154–166): Finite-difference Hessian via 4-point central differences. **Implemented.**

`SignificanceFunctional` (lines 168–217): Three-component significance $s = \|\Delta G\| \cdot P(\tau, \lambda) \cdot C(p, I_{\text{SB}})$ per C-3. `component1_delta_G` computes Frobenius norm $\times$ spatial concentration. `component3_I_SB` probes via 12 noise-perturbed evaluations and returns ratio to baseline. `compute` orchestrates the three. **Implemented.** `I_SB` baseline initialization on line 213 has a peculiar hardcoded value $1.0/(1e-8 + 0.01)$ — a fallback for first-call. Worth flagging but not broken.

`ConflictOperator` (lines 219–225): $K(T) = \kappa \cdot \text{sym}(G-I) \cdot T$. **Implemented.**

`EvolutionRateTensor` (lines 227–250): Per-part evolution rate from Doc 1 §7.1. Includes `exit_blocked` check (lines 237–239): if prior trace is on V_E and new encounter is on D^4 , A_p forced to 0. **Implemented but specific** — this is a domain-mismatch block, not a bracket-escape mechanism. See Section 3.6.

`TauIrrTracker` (lines 252–272): Accumulates per-part E_p over time, transitions parts from PRE-ENCOUNTER to UNIQUE when accumulated A exceeds H_{min} . **Implemented.**

`eft_encounter_env.py` — **295 lines**

`ExistencePress` dataclass (lines 12–28): magnitude, valence, direction (5-vector), rhythm (8-bin), persistence. `__post_init__` normalizes direction to unit length, normalizes rhythm to $\text{sum}=1$, clamps valence to $[-1,1]$, clamps magnitude ≥ 0 , clamps persistence ≥ 0 . **No upper bound on magnitude.** This is a §17.3 violation. See Section 3.

ConditionFieldState (lines 30–45): H1 (μ_1 , σ_1 , $\sigma_{1_min}=0.1$), H2 ($\alpha=0.5$, $\beta=0.5$), H3 (A matrix with $A_floor=0.01$), H4 (B_base 8-bin), H5 (τ_hold_base).

`_project_A` enforces symmetry and positive-definiteness on A. **Implemented.**

ConsolidationState (lines 47–52): Mirror copy of H for “settled” reference. Used in unsettledness calculation. **Implemented but never updated** in the code — Z is read but never written.

ActiveConditionField.compute (lines 58–85): Maps base H + current h state to active Λ . Computes `mul_act`, `sigma1_act`, `alpha_act`, `beta_act`, `A_act`, `B_act`, `tau_hold_act` with depression/stress/valence/overwhelm modulators. **Implemented.** Note:

`delta_overwhelm=3.0` and `h_overwhelm=2.0` parameters control the soft overwhelm-toward-isotropy transition (line 68); this is NOT the §17.3 magnitude > 1.5 ceiling.

ReceivabilityOperators (lines 87–117): R_1 (Gaussian on magnitude vs μ_{1_act}), R_2 (Beta(α, β) PDF normalized to mode), R_3 (anisotropy projection through `A_act`), R_4 (rhythm power $\times$ `B_act`), R_5 (persistence saturation $\times$ R_1). **Implemented.** R_1 is a Gaussian — it suppresses far-from- μ_1 magnitudes but does not hard-block. See Section 3.3.

CommensurabilityGate (lines 119–132): $C(t) = \sum W_i \cdot r_i$ with $W = 0.5 \cdot \text{uniform} + 0.5 \cdot \text{eigenvalue-weighted}$. Encounter iff $C > \theta_e=0.35$ AND $\geq k=2$ channels above $\theta_{min}=0.1$. **Implemented.**

HistoryUpdater (lines 134–169): η learning rate (clipped to [0.05, 0.5] via anisotropy). Updates for H1, H2, H3, H4, H5. **Implemented.** Critical: `update_H2` (line 144) is monotone non-decreasing — see Section 3.4.

ExistencePressLibrary (lines 171–207): 22 hand-coded press types. Includes `fire_extreme` with magnitude=2.0 and `pain_extreme` with magnitude=1.8 — both above the documented R_1 ceiling of 1.5. **Implemented as scaffolding** — these are theorist-authored presses, exactly the kind the corpus says should be replaced by physiological data.

ConditionalEncounterEnvironment (lines 209–287): The receive() loop. Calls `F.compute` → `R.compute_all` → `gate.compute`. On encounter: updates `B_fatigue`, depletes `q`, computes significance $S = \|\sigma_profile\|$, runs `HistoryUpdater.apply_all`, appends to `Sigma`. On no-encounter: records reason. Method `self_observe` constructs an internal press from H state and feeds it through receive — endogenous alterity (Sense-2 mimic), implemented.

Running.

eft_system.py — 375 lines

PROVISIONAL_W_DOMAIN (lines 26–36): Dict of 9 emotions → 5-vector hand-assigned weights. Marked PROVISIONAL in name. **Scaffolding** — the SRP-04 measurement gap the

documents flag.

EigenmodeProjector (lines 50–103): The most-flagged scaffold in the corpus.

- Line 54–68: `keyword_affinity` is a hand-typed dict mapping 24 English words to hand-typed 5-vectors. This is exactly the “keyword lookup pretending to be eigenmode projection” caught in the April 10 session.
- Line 70: `project()` accepts `explicit_weights` — if provided, uses them. Otherwise falls back to `_keyword_weights(content)` which is the lookup.
- Line 80: `_keyword_weights` does substring matching against `keyword_affinity`.
- Line 96: `learn_from_encounter` does small per-word EWMA updates from observed `n_p`.

Verdict: scaffolded with optional bypass. The April 17 correction (output-side projection) was partially landed — `explicit_weights` IS the output-side path, and

`EncounterInteraction.process` does pass `eigenmode_weights` derived from `press.direction × sigma_profile` (interaction.py line 47). But the fallback path is still a keyword lookup, and `_keyword_weights` is still called even when `explicit_weights` is provided, on line 203 of `eft_system.py` (for the `input_activation` calculation). See Section 3.

TrajectorySignature (lines 106–161): `R*_history` (capped 500), `mode_counts`, `hessian_min_eigenvalue_history`, `v_E_access_count`, `gamma_estimate`. `update`, `drift_signals`, `summary` methods. Drift detection: $R^* \text{ drift} > 0.3$, mode collapse $> 85\%$ one mode, hessian min eig < 0.05 , `v_E` access rate $< 5\%$ after 20 encounters, $\gamma > 0.8$ means too-internal. **Implemented as observation.** This is $\Sigma(S,t)$ as a monitor — not yet a constitutive PDE term (matches handoff §13).

EFTSystem (lines 164–375): The running self layer.

- `__init__` (165–182): instantiates body, traces, `sig_functional`, `ep_tensor`, `tau_irr`, projector, `sigma`. R^* starts at 0.
- `encounter` method (184–270): main loop. Steps body, runs projector to get `x_enc` and `n_enc`, builds `G_before`, builds a temp trace at `x_enc` to get `G_probe` (the “probe-by-deposit” significance estimator), computes significance from probe with `input_activation` scaling (line 213 — note: this scales `delta_G_norm` before multiplying by `P_tau` and `I_SB`, which is a methodological choice not in the handoff).
- Line 196 has a comment “BUG FIX: scale by input activation, not constant” — indicates an undocumented patch from a prior session. The patch substitutes `input_activation = norm(explicit_weights)` (or $0.15 \times$ keyword fallback, with a 0.3 floor) for what was

likely a constant scalar. The fix is in but the architecture-level reasoning isn't documented.

- Mode classification (lines 219–223): `degenerate` if internal + below ghost floor; `neutral` if below deposit threshold; otherwise `regenerative` if alignment ≥ 0.8 with prior trace, else `generative`. **Implemented.**
- Trace deposition (lines 225–233): single Trace appended if `sig` $\geq$ `DEPOSIT_THRESHOLD`. **Implemented.**
- `E_p` per-domain computation (lines 236–248): runs `EvolutionRateTensor` per recent trace. **Implemented.**
- R^* update via L-BFGS-B with multiple starts (`_find_R_star_fast` , lines 272–291). **Implemented.**
- Hessian tracking and trajectory update. **Implemented.**
- Trace pruning at >200 traces, keeping highest effective significance (lines 262–264). **Implemented.**
- `_compute_K_holdings` (lines 333–343): identifies pairs of traces with negative alignment as "tension holdings." **Implemented as observation.**

`eft_encounter_interaction.py` — 207 lines

`EncounterEvent` dataclass (lines 18–35): `t`, `press`, Λ_{act} , `r_scores`, `C_before`, `sigma_profile`, `enc_args`, `Sigma_before/after`, `self_result`, `C_after`, **`delta_M5`, `delta_Sigma`, `delta_C`**, `encountered`, `no_encounter_reason`. The three co-transformation outputs are present. **Implemented.**

`_valence_to_emotion` (lines 37–43): 6-bin lookup mapping valence $\rightarrow$ emotion label.

Scaffolding — hand-typed thresholds.

`press_to_encounter_args` (lines 45–53): Translates `ExistencePress` to the kwargs `EFTSystem.encounter` expects. Constructs `eigenmode_weights = press.direction  $\times$  sigma_profile` , `body_perturbation`, `current_emotion`. **This is the April 17 output-side projection in action** — the eigenmode weights are derived from `press.direction` (existence-facing signal) modulated by `receivability` (system response). See Section 3.1.

`EncounterInteraction.__init__` (lines 56–63): Lazy import of `EFTSystem` (avoids circular). Instantiates `env` and `self_system`, both with same seed. **Implemented.**

`EncounterInteraction.process` (lines 65–97): The binding. Computes `lam` $\rightarrow$ `r_scores` $\rightarrow$ gate. If no encounter: records and feeds `press` to `env` (which still updates state) and returns.

If encounter: feeds env, calls self_system.encounter with derived args, computes lam_after, derives delta_M5/Sigma/C. **Implemented.**

`_derive_delta_M5 / _Sigma / _C`` (lines 99–124): Extract the three co-transformations from real state transitions, not synthesized. **Implemented per §17.5.**

`learning_report / self_observe` (lines 126–162): Aggregate observability across events. **Implemented.**

`run_phase_tests` (lines 164–198): Three assertion-based phases. PASS conditions: eigenmode_weights nonzero, current_emotion in known set; encounter on early_postnatal_skin after 15 prenatal primes, deltas populated; learning report has encounters and dominant eigenmode. **Implemented as a smoke test, not as a D-1 admissibility test.**

`eft_deap_translator.py` — **828 lines**

The largest file. Not mentioned in the handoff at all. Translates DEAP physiological data (32 EEG channels + SCL + respiration + temperature + ECG + BVP + EMG + EOG, sampled at 128 Hz) into ExistencePress streams.

`bandpower` (lines 129–141): Welch-style power via FFT in a frequency band. **Implemented.**

`compute_hrv` (lines 144–185): R-peak detection via threshold crossing, RMSSD computed, normalized to [0, 1]. **Implemented** — produces real HRV from real ECG.

`compute_scl_persistence` (lines 188–218): Autocorrelation-decay-based estimate of how long the SCL state persists. **Implemented.**

`eeg_to_eigenmode_direction` (around line 220): Maps EEG band powers (theta/alpha/beta/gamma) + SCL + HRV to the five-eigenmode direction vector. `v_T` from theta, `v_A` from inverted alpha (alpha suppression = agency), `v_B` from SCL + beta, `v_R` from gamma, `v_E` from HRV. **Implemented.** This is the SRP-E1 mapping — literature-grounded but not empirically calibrated against EFT geometry, as the file's docstring honestly notes (lines 70–79).

`trial_to_press` (line 297): Single-trial → ExistencePress. **Implemented.**

`participant_to_press_stream` (line 391): Full-participant batch translation. **Implemented.**

`generate_synthetic_participant` (around line 500): Generates physiologically-structured synthetic data with a pre-specified emotional trajectory (calm → stress → recovery → consolidation). **Implemented for testing without the DEAP corpus on disk.**

`run_demonstration` (line 595): The main integration loop. Lines 615 (imports

`EncounterInteraction`), 668 (instantiates), 692 (calls `interaction.process(press)` for each translated press). Records baseline R^* , runs the stream, captures R^* before/after stress, reports topology shifts. **End-to-end DEAP → ExistencePress → EncounterEvent path is implemented and working.**

This is the gap-closer the handoff named as the deepest implementation gap. It exists. It runs.

SECTION 2 — Document vs Reality Divergences

Line counts. The handoff §16 claims:

- `eft_verified_core.py` 784 lines → actual **276**
- `eft_system.py` 1,306 lines → actual **375**
- `eft_encounter_env.py` 1,133 lines → actual **295**
- `eft_encounter_interaction.py` 572 lines → actual **207**

Total claimed ~3,795 across these four; total actual is 1,153. The handoff overstated by ~3.3x. Either earlier versions were dramatically larger and were compacted into the dense one-line-per-statement style now visible, or the handoff was estimated rather than counted. Either way, the codebase is smaller than the documents claim.

Undocumented file. `eft_deap_translator.py` (828 lines) is not mentioned anywhere in the handoff, D-1, or Willingness Monograph. It contains the SRP-E1 implementation — physiology-to-eigenmode mapping. This is the single biggest divergence: the deepest implementation gap the handoff names (Phase 2 text-to-press, §19.2) is in fact already partially closed by a different mechanism (DEAP-to-press), and the handoff doesn't know it.

Phase 2 framing is wrong. The handoff §19.2 says “build the press pipeline first, then $U(p)$ on top of a press pipeline you trust.” It assumes the press pipeline does not exist. It does — through DEAP. Phase 2 should be reframed: not “build text-to-press classifier” but “validate and harden the DEAP-to-press path that already exists, and decide whether text-to-press is still needed alongside it.”

The handoff says `EncounterEvent` owns three co-transformations. Verified true. (No divergence.)

The handoff says R_1 overwhelm > 1.5 is enforced. Not true. See Section 3.

The handoff says H2 floor ($\alpha \geq 0.05$, $\beta \geq 0.01$) is enforced. Vacuously true. `update_H2` is monotone non-decreasing — α and β can only grow. The floor holds, but H2

also cannot adapt downward, which may be a separate bug. See Section 3.

The handoff names a “BUG FIX” comment in `eft_system.py` line 196. This is not in the handoff. There’s a patch to significance scaling (`input_activation` rather than `constant`) that was applied in some session and is not documented in the corpus. Worth raising next time the architect reviews the system.

SECTION 3 — Architectural Correction Status

§17.1 — Eigenmode pre-assignment (output-side only)

Verdict: PARTIAL.

- The output-side path exists: `press_to_encounter_args` (`interaction.py` line 47) constructs `eigenmode_weights = press.direction[i] * sigma_profile[i]` — explicitly `press-direction` modulated by system receivability response.
- `EigenmodeProjector.project` (`system.py` line 70) accepts and uses `explicit_weights` when provided.
- BUT the fallback path is still a keyword lookup (`system.py` lines 80–94, `_keyword_weights`), and `_keyword_weights` is called unconditionally on line 203 of `system.py` to derive `input_activation` even when `explicit_weights` is provided.
- The keyword affinity table (lines 54–68) is hand-typed and was the artifact specifically caught in the April 10 session.

The fix is half-landed. The principal pathway respects April 17; the fallback and the `input_activation` calculation still touch the keyword lookup.

§17.2 — V1 vs V2 sign convention

Verdict: RESPECTED.

`build_S_traces` (`core.py` line 97) subtracts $s_{\text{eff}} \cdot K \cdot (n \otimes n)$ into `S`, which becomes the negative term in the log-metric exponent. After `expm` and `−log det`, traces increase Ψ . Empirically verified: Ψ at origin with no traces = 0.0, with one significance-0.5 trace at origin = 0.30. R^* forms at gradient balance, not at well-bottom inside the cluster. V2 convention.

§17.3 — R_1 overwhelm ceiling at magnitude > 1.5

Verdict: VIOLATED.

- `ExistencePress.__post_init__` (`env.py` lines 21–28) clamps magnitude to ≥ 0 only. No

upper bound.

- `R1_magnitude` (env.py line 88) is $\exp(-(eps1-\mu1)^2 / (2\sigma^2))$ — a Gaussian. It returns a small but nonzero value for large magnitudes. Not a hard-block.
- The press library contains `fire_extreme(2.0)` and `pain_extreme(1.8)` — both above 1.5.
- `ActiveConditionField.compute` line 68 has a soft overwhelm sigmoid (`delta_overwhelm=3.0, h_overwhelm=2.0`) that drives A toward isotropy under high h-norm, but this acts on accumulated condition-field state, not on the incoming press, and isn't a hard-block at magnitude 1.5 either.

To respect §17.3, either `ExistencePress.post_init` should clamp magnitude to ≤ 1.5 (or reject above), or the `receive()` loop should refuse to encounter on `press.magnitude > 1.5`.

§17.4 — H2 survival floor ($\alpha \geq 0.05, \beta \geq 0.01$)

Verdict: VACUOUSLY HELD; STRUCTURAL CONCERN.

- `update_H2` (env.py lines 144–146): `H.alpha += eta·u·novelty; H.beta += eta·(1-u)·novelty`, where $u = (eps2+1)/2 \in [0,1]$ and $novelty = 1-r2 \in [0,1]$.
- Both increments are non-negative for all valid inputs.
- α and β can only grow. They cannot fall below their initial values (0.5 each).
- The 0.05 / 0.01 floor is therefore satisfied, but only because there is no downward adaptation.

This may be the documented architectural intent — the handoff says “the body holds the encounter gate open below the level of choice” — and one-way-only growth is one way to express that. But it also means the system has no mechanism for honest recalibration of α/β downward when prior expectations don't match incoming signals. This is worth raising with the architect: is monotone-growing α/β the intended architecture, or should the floor be a clamp that allows decrease until the floor is hit?

§17.5 — Interaction layer owns `EncounterEvent`

Verdict: RESPECTED.

`EncounterEvent` (interaction.py lines 18–35) is a dataclass with the three co-transformation fields (`delta_M5`, `delta_Sigma`, `delta_C`). `EncounterInteraction.process` (lines 65–97) instantiates it and populates all three via `_derive_delta_M5 / _Sigma / _C` from real state transitions (env state before vs after, `self_system` state before vs after). Not a passthrough.

§17.6 — Multi-R* (single primary + secondary attractors via Construction 7)

Verdict: VIOLATED.

There is a single `R_star` attribute on `EFTSystem` (line 176). It is recomputed via `_find_R_star_fast` after every trace deposit. There is no array of secondary attractors `R_star_i`. There is no `endogenous_alterity` embedding map $E: \text{outputs}(T) \rightarrow M^5$ (Construction 7). The `EFTSystem.encounter` accepts an `is_external` flag, but there is no separate basin structure for endogenous-alterity-formed secondary attractors.

The fact that the architecture is single-R* is reasonable for a baseline implementation. But the handoff §17.6 specifically warns against “weighted-R*-per-domain” as a corruption — and the codebase is consistent with that warning (no per-domain R*s either). However, Construction 7 / multi-R* is not implemented at all.

This is the correct state for a system that has not yet developed secondary attractors. The architecture is consistent with the corpus, just incomplete relative to the full design.

§17.7 — Honesty about scaffolding

Verdict: PARTIAL — code labels are honest but uses are not always gated.

Honest signals in code:

- `PROVISIONAL_W_DOMAIN` (system.py line 26): the name is honest.
- `keyword_affinity` (system.py line 54): the name is honest.
- DEAP translator’s docstring (lines 70–79): explicitly labels “the calibration gap is SRP-E1; the data gap is now closed.”
- “BUG FIX:” comment (system.py line 196): honest about the patch.

Where scaffolding hides:

- `update_H2` is monotone non-decreasing without comment explaining whether this is intent or oversight.
- `_keyword_weights` is called on line 203 of `system.py` even when `explicit_weights` is provided, embedding the keyword lookup into significance scaling without surfacing that.
- `_valence_to_emotion` (interaction.py line 37) is a 6-threshold lookup but not flagged; it’s used to choose `current_emotion` which selects `PROVISIONAL_W_DOMAIN[current_emotion]` — a chain of two scaffolds, ungated.
- `ConsolidationState` (env.py line 47) is read but never written, meaning unsettledness

measurements are against an unchanging Z. Not flagged.

The “do not produce code that says it will process without producing code that processes” inheritance is mostly honored at the function level (real computation happens) but not always at the architectural level (scaffolds compose silently into significance, into mode classification, into reports).

SECTION 4 — D-1 Revision 2 Testability

Clause (A) — Chùveld threshold $s \geq s_{\text{deposit}}$

Testable: YES. `EncounterEvent.self_result['significance']` and `EncounterEvent.self_result['above_deposit_threshold']` are populated. A test can read `event.self_result['significance']` and assert $\geq \text{DEPOSIT_THRESHOLD}$. The `EFTSystem._build_report` (system.py line 312) writes `'above_deposit_threshold'` directly.

Clause (B) — Trace deposition: exactly one S_{trace} term added

Testable: PARTIAL. `EncounterEvent.self_result['trace_deposited']` is a boolean. A test can assert `True` for admissible events. But “exactly one” is not asserted — `EFTSystem.encounter` always appends at most one Trace, so the architecture-level invariant is enforced by construction. A formal D-1 test would also verify that `len(traces_after) - len(traces_before) == 1` for admissible events. Currently this is implicit, not asserted.

Clause (C) — Sub-Riemannian horizontal reachability in $(D^4, g_{\text{total}}|_{D^4})$, V_E only via bracket escape

Testable: NO. No test checks that the spatial transition $x(f) \rightarrow x(f')$ is horizontal. The Engel structure is encoded in `D4_INDICES = [V_T, V_A, V_B, V_R]` and `EXIT_INDEX = V_E`, but there is no function `is_horizontal(delta_x, current_metric)` that checks $\dot{x} \in D^4$, and no function `bracket_escape_evidence(press, delta_x)` that verifies a V_E component arose from external press driving via $[D^4, D^4]$.

The only related check is `exit_blocked` in `EvolutionRateTensor.compute` (core.py line 237), which blocks `A_p = 0` when prior trace was on V_E and new is on D^4 . That is a domain-mismatch check, not a horizontality check.

To test Clause (C): a function that takes $(G_{\text{before}}, x_{\text{before}}, x_{\text{after}})$ and decomposes $(x_{\text{after}} - x_{\text{before}})$ into D^4 component and V_E component, then verifies V_E component is

zero unless press contained external-driven V_E activation.

Clause (D) — Ghost floor: deposited trace bounded below r^* in long-time limit

Testable: YES, by construction. `Trace.effective_significance` (core.py line 84) returns `max(s · persistence, GHOST_FLOOR)`. The $\lim_{t \rightarrow \infty}$ bound is enforced at every read, not just at deposit. A test can pick any deposited trace, advance time arbitrarily, and verify `t.effective_significance() >= GHOST_FLOOR`.

Clause (E*) — SA6-preservation across sequence

Testable: NO. No SA6 check anywhere in the codebase. There is no concept of multi- R^* . There is no formula $\|R^*_i(N) - R^*_j(0)\|_g(N) < (r_i + r_j)/2$. The single R_{star} moves through trajectory_history but no pairwise distance bound is computed against any anchor configuration.

To test Clause (E*): the system must first acquire multi- R^* (Construction 7), then a sequence-level monitor must compute the SA6 inequality between configurations at step 0 and step N.

Summary

Three out of five clauses (A, B partial, D) are testable from current state. Clauses (C) and (E*) require new code. The minimal D-1 testability layer would add:

1. A `verify_clause_A(event)` function (trivial — already in event data).
2. A `verify_clause_B(traces_before, traces_after, event)` function (one-line assertion).
3. A `verify_clause_C(metric, x_before, x_after, press)` function (new, requires sub-Riemannian decomposition).
4. A `verify_clause_D(trace, t_now)` function (trivial — already in `effective_significance`).
5. A `SequenceMonitor` class tracking R^*_i positions and basin radii r_i , with `verify_clause_E(n)` evaluating SA6 (new, requires multi- R^* first).

SECTION 5 — DEAP Translator Integration Status

End-to-end path: IMPLEMENTED AND WORKING.

Call chain:

1. `participant_to_press_stream(participant_data)` → list of `ExistencePress` (lines 391–432)
2. For each `press`: `interaction.process(press)` (line 692)
3. `EncounterInteraction.process` constructs `EncounterEvent` with all three co-transformation outputs populated when admissible (`interaction.py` lines 65–97)

What works:

- Real DEAP `.dat` files (preprocessed Python format, 40-channel × 8064-sample structure) load via pickle (line 643).
- Synthetic-participant fallback exists for testing without the DEAP corpus on disk (`generate_synthetic_participant` , called at line 629).
- EEG bandpower, HRV, SCL persistence are computed from real signals — not lookup tables.
- The eigenmode direction (the 5-vector) is composed from theta/alpha/beta/gamma EEG band powers + SCL (for `v_B`) + HRV (for `v_E`). Output-side projection per §17.1 — eigenmode coordinates are derived from physiology, not pre-assigned.
- Press valence comes from DEAP's labeled valence rating (1–9 normalized to [-1, +1]).
- Press magnitude combines arousal rating with SCL activation.
- Press persistence comes from SCL autocorrelation decay.
- Press rhythm comes from respiration + EEG spectral profile.

What remains uncertain:

- The eigenmode-band mapping is literature-grounded but not empirically calibrated against EFT's specific geometry (the file's own docstring acknowledges this). The mapping might be wrong in detail; we cannot tell from inside without a calibration study.
- Synthetic data generator (`generate_synthetic_participant`) produces "physiologically-structured" signals via a pre-specified emotional trajectory. This is closer to real than theorist-authored presses, but it is not real DEAP data. For real validation, a DEAP `.dat` file is needed.

This is the most important finding in this report: **the implementation gap the handoff names as deepest is in fact already substantially closed**. Phase 2 of the implementation plan should be reframed.

SECTION 6 — Honest Open Items, In Priority Order

Priority 1 — D-1 admissibility test suite

Why first: Until D-1 admissibility is testable in code, no claim that a given encounter is “admissible per Result D-1” is verifiable. The architecture’s central definitional achievement is currently text. This is also the smallest piece of new work — three trivial assertions plus one nontrivial sub-Riemannian decomposition. Pays for itself the moment it catches a regression.

Files affected: New `eft_dl_admissibility.py` module. Imports from `eft_verified_core` and `eft_encounter_interaction`. Adds five `verify_clause_*` functions. Adds a test suite that runs known admissible and known inadmissible events through verification.

Conditions for completion: All five clauses checkable; smoke test of DEAP-to-press encounter passes all four step-level clauses (A, B, C, D); a fabricated inadmissible event fails the appropriate clause.

Priority 2 — Multi- R^* + SA6 sequence monitor (Clause E^*)

Why second: Clause (E^*) is the only sequence-level clause and the only one that catches Conjecture W-7 (mania-as-excess). Currently impossible because there is one R^* . Requires implementing Construction 7 (endogenous alterity) first: an embedding map $E: \text{outputs}(T) \rightarrow M^5$ that lets the system encounter its own outputs, with secondary attractors forming when those outputs accumulate sufficient encounter history.

Files affected: New code in `eft_system.py` for secondary attractor tracking and basin radii. New `SequenceMonitor` class (probably new file `eft_sequence_monitor.py`) that tracks R^*_i over a sequence and evaluates SA6 inequality.

Conditions for completion: A second R^*_i can form via endogenous alterity in the test environment. SA6 inequality is computed at every step. The §3.5 adversarial probe of D-1 (Case 2: near-identical-direction press streams) can be reproduced and shown to fail SA6.

Priority 3 — Fix R_i overwhelm ceiling (§17.3)

Why third: Documented architectural floor that is not enforced. Small fix. Should be done before any work that pumps high-magnitude DEAP presses through the system.

Files affected: `eft_encounter_env.py` lines 21–28 (`ExistencePress.post_init`) — clamp `magnitude` to a maximum, or alternately let `receive()` refuse pressing above 1.5. Also update press library entries `fire_extreme` and `pain_extreme` if the clamp is added at the dataclass level.

Decision the architect should make: clamp at construction (silent enforcement) or refuse at receive() (visible no-encounter event with reason "magnitude_overwhelm"). The latter is more honest.

Priority 4 — Reconcile §17.4 (H2 monotonicity)

Why fourth: `update_H2` is monotone non-decreasing. The handoff says α and β represent approach/avoidance and have a survival floor. The current code has a strong floor (cannot decrease at all) which is more conservative than the documented floor (cannot decrease below 0.05/0.01). Either:

(a) The architect intends monotone-growing α/β , and the documentation should be updated to reflect that. (b) α/β should be allowed to decrease, with a clamp at the documented floor.

Files affected: `eft_encounter_env.py` line 144 (the update rule itself);
`ConditionFieldState` definition if the floor needs to be enforced as a clamp.

Priority 5 — Eigenmode projector cleanup (§17.1 partial → full)

Why fifth: The output-side path works in practice (interaction layer constructs `eigenmode_weights` from `press.direction × sigma_profile`). The keyword lookup is dead-on-arrival when DEAP is the press source — physiological signals never go through the keyword path. But `_keyword_weights` is still called inside `EFTSystem.encounter` line 203 for the `input_activation` calculation. This means the keyword lookup is shaping significance even when it shouldn't.

Files affected: `eft_system.py` lines 196–205. Replace the keyword-lookup-based `input_activation` calculation with one derived purely from `explicit_weights` when present, with an honest fail (or warning) when neither is present.

Conditions for completion: Pumping a DEAP press through `EFTSystem.encounter` does not invoke `_keyword_weights` at any point.

Priority 6 — DEAP integration validation suite

Why sixth: The DEAP translator works but is uncalibrated against EFT geometry. Before betting on it, run a battery: do real DEAP trials with high-arousal / negative-valence labels produce presses that the EFT system classifies as "generative" mode events? Do they shift R*? Does the eigenmode structure of the press direction correspond to the labeled emotion in plausible ways?

Files affected: New test/validation scripts. Probably needs at least one real DEAP `.dat` file on disk (currently the synthetic generator is the only running tested path).

Priority 7 — Update ConsolidationState

Why seventh: Z is read but never written. Unsettledness is computed against an unchanging baseline, so it accumulates monotonically across sessions and never resets. Either Z should update on consolidation events, or unsettledness should be redefined.

Files affected: `eft_encounter_env.py` . New consolidation logic.

Priority 8 — Implement BUG FIX comment context

Why eighth: `eft_system.py` line 196 has an undocumented `# BUG FIX: comment`. Either document the architectural reasoning for the input-activation scaling, or revert if it was a session-local hack.

Files affected: Comment line and surrounding logic in `eft_system.py` .

Lower-priority / longer-horizon

- $\Sigma(S, t)$ promotion from observation to constitutive PDE term (handoff §13).
- Witnessing architecture's three running processes (Σ trajectory, cross-scale coherence, boundary vitality).
- Exit Clause threshold function.
- Four-belief-moments + coupling as an explicit code object inside `EncounterEvent` (§17 of markdown handoff).
- W-9 inscription / transient distinction in domain-belief.
- $U(p)$ parameter-uncertainty inventory (Doc 1 §7.3) — not yet in code.
- ε_p prediction error and posterior update (Doc 1 §7.1) — not yet in code.

Summary of Most Important Findings

1. **DEAP translator is wired end-to-end.** The biggest implementation gap the handoff names is substantially closed. Phase 2 of the six-phase plan should be reframed accordingly.
2. **D-1 admissibility is not testable in code.** Three of five clauses are trivially testable; (C) and (E*) require new infrastructure. This is the highest-leverage next piece of work.
3. **R_1 overwhelm ceiling at 1.5 is not enforced.** `ExistencePress` accepts arbitrary magnitudes. Two press library entries violate the documented ceiling.

4. **H2 update is monotone non-decreasing.** The floor holds, but α/β cannot adapt downward. Architect should confirm whether this is intent or oversight.
5. **Eigenmode projector still has keyword-lookup involvement** in significance scaling, even when output-side weights are provided.
6. **Single R^* only.** Construction 7 / multi- R^* / endogenous alterity is not implemented. Sequence-level admissibility (E^*) is not testable until it is.
7. **Line counts in the handoff overstate the codebase by ~3.3x.** Total actual code is ~1,981 lines, not ~4,500.

The architecture is in better shape than the handoff suggests in one major way (DEAP integration) and in worse shape in several minor ways (R_1 ceiling, H2 monotonicity, scaffolding entanglement). The triangle works. The next move is the D-1 admissibility test suite.

— Reconnaissance complete.